Sre Cibi Varsshan K

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EDUCATION

Vellore Institute of Technology, Chennai | Expected 2027
B.Tech Computer Science and Engineering (Artificial Intelligence and Machine Learning)
Relevant coursework: Data Structures and Algorithms, Computer Graphics, Machine Learning, Computer Vision.

SUMMARY

Computer science student building native software: a C++17 video tool on the FFmpeg C API, an offline Android LLM app with a JNI bridge to llama.cpp, verified Linux install tooling, GPU compute shaders and a from-scratch TD3 agent.

PROJECTS

censface-cpp | C++17, FFmpeg, OpenCV, CMake, Linux | Oct 2026
github.com/xerfzy/censface-cpp

- Built a Linux command-line tool that demuxes and decodes video with libavformat and libavcodec, converts frames with libswscale, runs OpenCV DNN head detection and effects, and encodes H.264 MP4 in a single pass.
- Preserved source timing by encoding in the input stream's time base with each frame's own timestamp; copied audio packets without re-encoding and applied the display-matrix rotation to pixels.
- Managed every FFmpeg object with RAII (unique_ptr with custom deleters); added atomic temp-file output, SIGINT cancellation and a batch script; verified 24 of 24 frame timestamps and byte-identical audio preserved on a test clip.

XrivenixAI - offline Android LLM assistant | Kotlin, Jetpack Compose, Room, JNI, C++ | Sep 2026
github.com/xerfzy/XrivenixAI

- Integrated llama.cpp through a C++ JNI bridge built with the Android NDK and CMake, with streaming generation and KV-cache reuse between turns.
- Designed a model-independent memory layer on Room/SQLite: FTS5 full-text search, embedding similarity and entity links merged into one score, plus validation that de-duplicates facts and versions them by topic.
- Wrote 30 JVM unit tests and 29 instrumented tests and benchmarks; measured about 5-6 tokens/s generation and about 1.8 s to first token with KV reuse (about 7 s without) for a 3B model on a Snapdragon 782G phone.

eos-oneplus-nord - verified Android OS install tooling | Bash, Linux, adb, fastboot | Oct 2026
github.com/xerfzy/eos-oneplus-nord

- Wrote Bash scripts that resume and SHA-256-verify a 2.1 GB image download, check device identity and firmware over adb, and audit the vendor installer against pinned hashes before anything runs.
- Gated the destructive flash on fastboot product and unlock state, a single attached device and a typed serial confirmation; used the tooling to install /e/OS on a OnePlus Nord.

ChaCha20 GPU Image Encryption | C++17, OpenGL 4.3, GLSL, OpenSSL | Mar - May 2026
github.com/xerfzy/chacha2020ImageEncryption_using_compute_shader

- Implemented the ChaCha20 block function in a GLSL compute shader; each invocation generates two keystream blocks and XORs them into a shader storage buffer (SSBO) holding a packed batch of images.
- Planned memory-bounded batches from image headers alone and implemented the upload, dispatch, memory barrier and mapped readback path with a Dear ImGui interface.

TrackMania TD3 - reinforcement learning research | Python, PyTorch, TMRL, pytest | Sep 2026
github.com/thenamelessone888/RL

- Implemented TD3 from scratch (twin critics, delayed policy updates, target policy smoothing) for TrackMania 2020 with LIDAR observations and a behaviour-cloned warm start from human driving.
- Diagnosed a post-warm-up policy collapse and fixed it with TD3+BC regularisation; traced a training plateau to a recurring wall collision from evaluation logs; covered the agent with 13 offline pytest tests.

RESEARCH

Indian Patent Application IN202641060337 A1 (published application, not granted) | May 2026
"A System for Swarm of Drone for Crowed Behaviour Analysis". Application No. 202641060337, filed 12 May 2026, published 22 May 2026. Inventors: Sre Cibi Varsshan K, D Kavitha. Applicant: Vellore Institute of Technology, Chennai.

TECHNICAL SKILLS

Languages: C++ (C++17), C, Kotlin, Python, C#, Bash, GLSL
Systems and tools: Linux, Git, CMake, GCC, Android NDK, adb, fastboot
Libraries and frameworks: FFmpeg (libavformat, libavcodec, libswscale), OpenCV, OpenGL, Jetpack Compose, Room/SQLite, llama.cpp, PyTorch, Unity
Practices: Unit and instrumented testing (JUnit, pytest), RAII resource management, checksum verification